Exercise for Sect. 2.7

Exercise **2.7.1**

Show that if a measure-preserving system $(X, \mathcal{B}, \mu, T)$ has the property that for any $A, B \in \mathcal{B}$ there exists N such that

$$\mu(A \cap T^{-n}B) = \mu(A)\mu(B)$$

for all $n \geq N$, then it is trivial in the sense that $\mu(A) = 0$ or 1 for every $A \in \mathcal{B}$.

Proof: □

Exercise 2.7.2

Show that if a measure-preserving system $(X, \mathcal{B}, \mu, T)$ has the property that

$$\mu(A \cap T^{-n}B) = \mu(A)\mu(B)$$

uniformly as $n \to \infty$ for every measruable $A \subseteq B \in \mathcal{B}$, then it is trivial in the sense that $\mu(A) = 0$ or 1 for every $A \in \mathcal{B}$.

Proof: $\square$

Exercise **2.7.3**

Asuume that $\mathcal{A}$ is a semi-algebra that generates $\mathcal{B}$, and prove the following characterizations of the basic mixng properties for a measure-preserving system $(X, \mathcal{B}, \mu, T)$:

1. T is mixing if and only if
$$\mu(A \cap T^{-n}B) \to \mu(A)\mu(B)$$
as $n \to \infty$ for all $A, B \in \mathcal{A}$.
2. T is weak-mixing if and only if
$$\frac{1}{N}\sum_{n=0}^{N-1} |\mu(A \cap T^{-n}B) - \mu(A)\mu(B)| \to 0$$
as $N \to \infty$ for all $A, B \in \mathcal{A}$.
3. T is ergodic if and only if
$$\frac{1}{N}\sum_{n=0}^{N-1} \mu(A \cap T^{-n}B) \to \mu(A)\mu(B)$$
as $N \to \infty$ for all $A, B \in \mathcal{A}$.

Proof: □

Exercise **2.7.4**

Let $\mathcal{A}$ be a generating semi-algebra in $\mathcal{B}$ andassume that for $A \in \mathcal{A}$, $\mu\left(A \Delta T^{-1} A\right) = 0$ implies $\mu(A) = 0$ or 1. Does it follow that T is ergodic?

Proof: $\square$

Exercise 2.7.5

Show htat a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is mixing if and only if

$$\lim_{n\to\infty} \langle U_T^n f, g\rangle = \langle f, 1\rangle\langle 1, g\rangle$$

for all f and g lying a dense subset of L^2_μ.

Proof: □

Exercise 2.7.6

Use exercise 2.7.5 to prove the following.

1. An ergodic automorphism of a compact abelian group is mixing with respect to Haar measure.
2. An erogidc automorphism of a compact abelian group is mixing of all orders with respect tot Haar measure.

Proof: □

Exercise 2.7.7

Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is weak-mixing if and only if

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \left| \langle U_T^n f, g\rangle - \langle f, 1\rangle\langle 1, g\rangle \right| = 0$$

for any $f, g \in L^2_\mu$.

Proof: □

Exercise **2.7.8**

Show that a measrue-preserving system $(X, \mathcal{B}, \mu, T)$ is weak-mixing if and only if

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} \left| \langle U_T^n f, f \rangle - \langle f, 1 \rangle \langle 1, f \rangle \right| = 0$$

for any $f \in L_\mu^2$.

Proof: □

Exercise **2.7.9**

Shoe that a Bernoulli shift is mixing of order k for every $k \geq 1$.

Proof: $\square$

Exercise **2.7.10**

Prove that a measure-preserving transformation T is mixing if and only if

$$\mu(A \cap T^{-n}A) \to \mu(A)^2$$

for all $A \in \mathcal{B}$. Deduce that T is mixing if and only if $\langle U_T^n f, f \rangle \to 0$ as $n \to \infty$ for all f in a set of function dense in the set of all L^2 functions of zero integral.

Proof: □

Exercise **2.7.11**

Prove that a measure-preserving transformation T is weak mixing if and only if for any measurable sets A, B, C with positive measrue, there exists soem $n \geq 1$ such that $T^{-n}A \cap B \neq \emptyset$ and $T^{-n}A \cap C \neq \emptyset$.

Proof: □

Exercise **2.7.12**

Write T^k for the k-fold Cartesian product $T \times \cdots \times T$. Prove that T^k is ergodic for all $k \geq 2$ if and only if T^2 is ergodic.

Proof: □

Exercise **2.7.13**

Let T be an ergodic endomorphism of $\mathbb{T}^d$. The following exponential error rate for the mixing property,

$$\left|\langle f_1, U_T^n f_2\rangle - \int f_1 \int f_2\right| \leq S(f_1)S(f_2)\theta^n$$

for some $\theta < 1$ depending on T and for the pair of constants $S(f_1), S(f_2)$ depending on $f_1, f_2 \in C^\infty(\mathbb{T}^d)$, is known to hold.

1. Prove an exponential rate of mixing for the map $T_n : \mathbb{T} \to \mathbb{T}$ defined by $T_n(x) = nx \bmod 1$.
2. Prove an exponential rate of mixing for the automorphism of $\mathbb{T}^2$ defined by $T : \binom{x}{y} \mapsto \binom{y}{x+y}$.
3. Could an exponential rate of mixing hold for all continuous functions?

Proof: □